

# STATUS EPILEPTICUS FIRST-30-MINUTE ALGORITHM

PGY2 call-night handout - verify local EEG/ICU pathways and order sets before use

## Definition: seizure $\geq 5$ min OR recurrent seizures without return to baseline.

Treatment and workup run in parallel. Do not wait for CT, labs, or EEG to treat active convulsive status.

### 0-5 MIN: STABILIZE

- Call for help; assign airway, access, meds, clock.
- Position, suction, oxygen as needed, monitors.
- Check glucose now; treat hypoglycemia.
- IV/IO access; labs and ASM levels when relevant.
- Pregnancy test when relevant; document onset time.

### 5-10 MIN: BENZODIAZEPINE

- Lorazepam 0.1 mg/kg IV, max 4 mg per dose.
- May repeat once after 3-5 minutes.
- No IV: midazolam 10 mg IM adult dose.
- Prepare second-line load immediately.
- Lorazepam 1 mg is usually underdosing.

### 20-40 MIN: FULL SECOND-LINE LOAD

- Levetiracetam 60 mg/kg IV, max 4.5 g.
- Valproate 40 mg/kg IV, max 3 g.
- Phenytoin 20 mg/kg IV or fosphenytoin 20 mg PE/kg IV, max 1.5 g / 1.5 g PE.
- Choose based on patient factors and local workflow.
- 1 g levetiracetam is not a status load for most adults.

### EEG TRIGGERS

- Persistent coma or altered mental status.
- Subtle eye deviation, facial twitching, nystagmoid movements.
- Fluctuating mental status or autonomic instability.
- Recent convulsive status epilepticus.
- Refractory status or ICU-level sedation.

### REFRACTORY SE: ESCALATE

- Ongoing seizures despite adequate benzodiazepine plus second-line ASM.
- Call ICU/neurocritical care/anesthesia early.
- Plan intubation, continuous EEG, anesthetic infusion per local protocol.
- Communicate exact timeline, doses, response, airway status.

### PARALLEL ETIOLOGY SEARCH

- Metabolic: glucose, Na, Ca, Mg, renal/hepatic failure.
- Infectious/structural: meningitis, encephalitis, stroke, hemorrhage, tumor, trauma.
- Toxicologic: withdrawal, INH, TCA, stimulants, bupropion, tramadol.
- Autoimmune/NORSE, pregnancy/eclampsia, nonadherence, low ASM levels.

### AFTER THE PATIENT IS SAFE: SEIZURE VS SYNCOPE VS FUNCTIONAL EVENTS

| Clue          | Epileptic seizure                                     | Syncope                                         | Functional events                                                     |
|---------------|-------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------|
| Before        | Aura; sleep deprivation; missed ASM                   | Standing, heat, pain, dehydration, skipped meal | Often situational or emotionally loaded, but diagnose positively      |
| During        | Eyes open/deviated; stereotyped synchronous movements | Pallor/diaphoresis; brief jerks near the end    | Forced eye closure; resists opening; asynchronous or waxing movements |
| After         | Lateral tongue bite; minutes to hours of confusion    | Rapid orientation; tip tongue bite possible     | Variable recovery; recall may be preserved                            |
| Must not miss | Status at $\geq 5$ min; glucose and cause search      | ECG belongs in transient loss of consciousness  | Real events; ASMs do not treat functional seizures alone              |

### SECOND-LINE STATUS LOADS: SAY THE WEIGHT-BASED DOSE OUT LOUD

| Drug         | Loading dose                                         | Maximum           | Practical note                                                        |
|--------------|------------------------------------------------------|-------------------|-----------------------------------------------------------------------|
| LEV          | Levetiracetam 60 mg/kg IV                            | 4.5 g             | 1 g is not a status load for most adults.                             |
| VPA          | Valproate 40 mg/kg IV                                | 3 g               | Avoid severe liver disease, thrombocytopenia, pregnancy/POLG concern. |
| PHT / fosPHT | Phenytoin 20 mg/kg IV or fosphenytoin 20 mg PE/kg IV | 1.5 g or 1.5 g PE | Use telemetry; avoid/think twice with conduction disease.             |

Core sources: ILAE operational definition; AES convulsive SE guideline; ESETT; ENLS SE protocol v6.0 if locally available.

Local placeholders to verify: EEG contact pathway, ICU escalation pathway, preferred second-line ASM order set.